

Treasure Island, FL — Hurricane Change Detection

A pre/post-hurricane comparison of Maxar, NOAA, and Sentinel-2 satellite imagery

Treasure Island, FL | 2023-11-04 to 2024-10-14 | Hurricanes Helene & Milton (2024)

Method

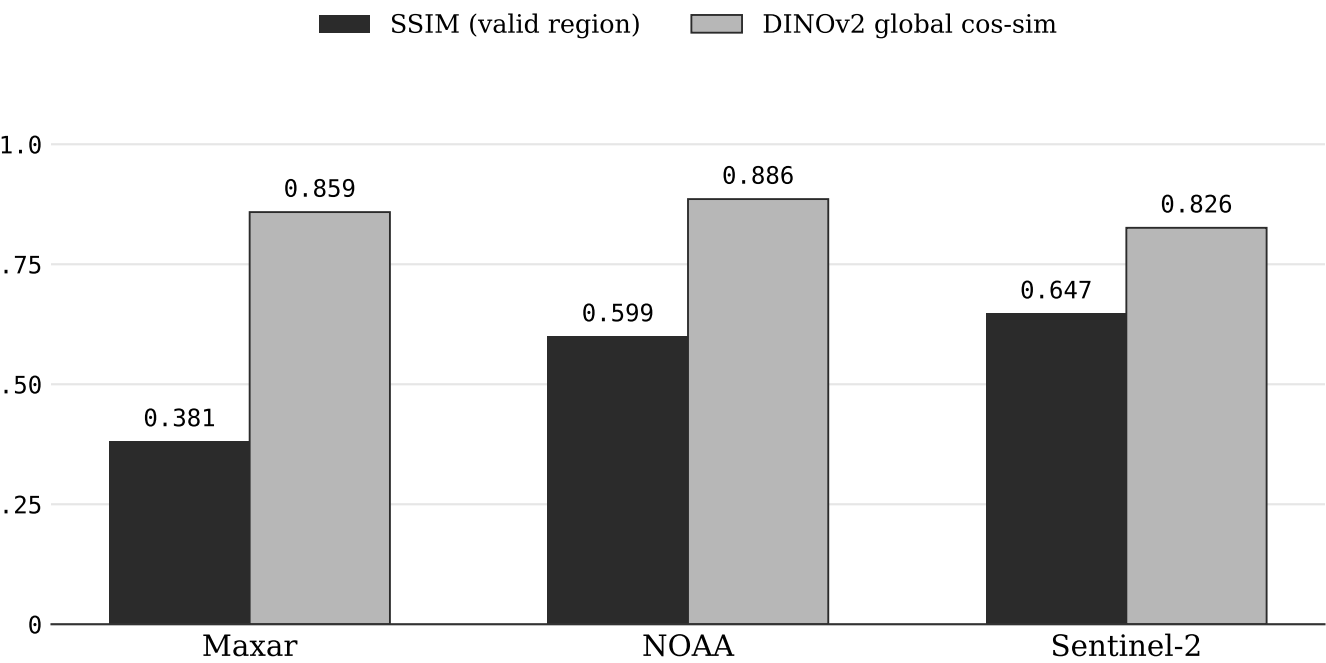
Each before/after pair is co-registered with ORB feature matching and a RANSAC homography, since the two dates rarely share an exact footprint. Two independent similarity signals are then computed over the registered pair: Structural Similarity (SSIM), a classical pixel/texture measure robust to lighting and tide differences, and DINOv2, a self-supervised vision transformer whose embeddings capture semantic content rather than raw pixel structure. An Otsu threshold on the SSIM map produces the change-mask overlays shown on page 2.

Results

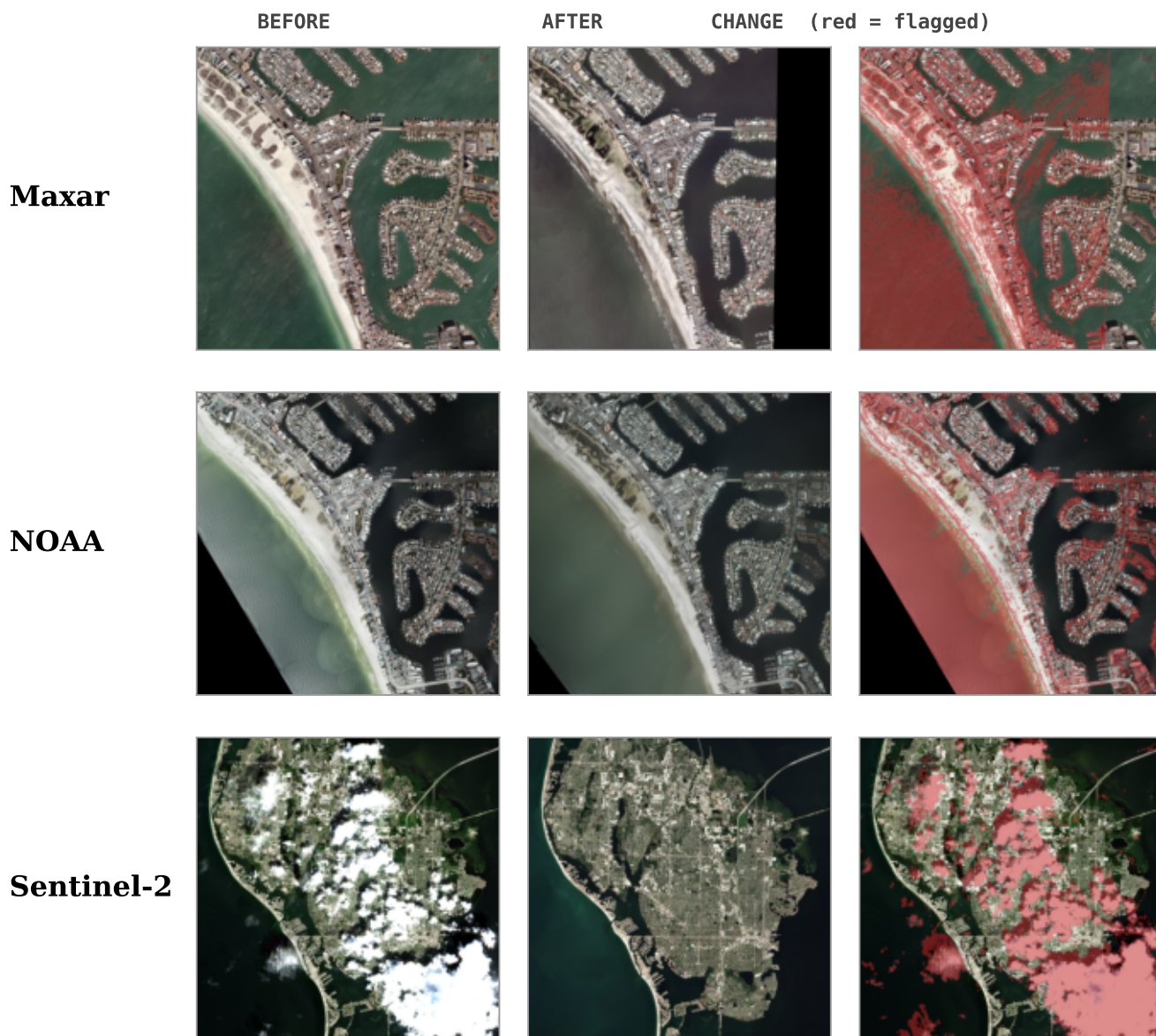
Pair	Dates	SSIM	% Changed	DINOv2 sim	Read
Maxar	20231104 to 20241010	0.381	54.0%	0.859	Severe shoreline change
NOAA	20240930 to 20241011	0.599	42.4%	0.886	Modest incremental change
Sentinel-2	20240919 to 20241014	0.647	28.0%	0.826	Cloud-confounded, use cautiously

Similarity by Method

1.0 = identical; higher = more similar



Visual Evidence — Before / After / Change



Conclusion

All three sources agree on the same basic story: the open Gulf-facing beach and dune line took substantial, visually confirmed damage between the pre-hurricane baseline and the post-Milton state, while interior canal neighborhoods stayed comparatively stable. NOAA's storm-on-storm pair shows Milton's incremental damage on top of Helene was real but smaller than the full pre/post change, and DINOv2 independently confirms it as the least-changed pair. The two methods disagree only on Sentinel-2: SSIM calls it most similar (cloud texture still reads as texture), DINOv2 calls it least similar (clouds break the semantic match) — a reminder that these percentages are relative indicators, not calibrated damage measurements.

Method: ORB+RANSAC registration -> masked SSIM / Otsu change mask (OpenCV, scikit-image) + DINOv2 (facebook/dinov2-base) cosine similarity.